CS-E4960 - Software Testing and Quality Assurance D, Lecture, 3.9.2024-26.11.2024

This course space end date is set to 26.11.2024 **Search Courses: CS-E4960**

Group assignments

Group registration
Done

Group name choice
To do

Analysing quality using ISO/IEC 25010
Done

Team testing
Done

Final report
To do

In this assignment, you will produce a small quality model, quality assurance plan, and test report as a group. This simulates defining and executing a quality assurance project in a real-life scenario, although you will not fully work out all the aspects of the plan.

To complete this assignment, do the following:

1. Read the project case description provided as part of this assignment.
2. Create a quality model according to ISO 25010 that concerns the information given in the project case description. You do not need to include every quality characteristic described in ISO 25010. Instead, you should first analyse what quality characteristics are relevant for the stakeholders listed in the project case description. You must include a table mapping user needs to quality characteristics, following the format described in Section 3.6 of ISO/IEC 25010 (primary, secondary, indirect user, etc.). You can reuse what you created for the Team testing assignment.
3. Write a concise description of the quality control activities (e.g., testing, inspection, reviews) that you plan to assure quality for the stakeholders. The description can be in table form and it should include information on the quality control activity, which qualities are assured by each activity, which stakeholder need is assured, and a short rationale for why this activity is suitable to assure the quality. This description must include software testing as a quality control activity.
4. Write a concise test plan to detail at least 10 test cases to ensure quality of the system. You can reuse tests that you have created earlier in this course for the Oscar system. There must be:
   - At least one automated unit test case.
   - At least one automated acceptance test case.
   - At least one manual test case.
5. Write a test report to detail the results of your test cases.
6. Execute at least the three tests listed in the previous point.
7. Summarise your assessment of the SUT's quality. Document your work in a single final report file.

More detailed information about these steps are provided in the project case description and final report template that are attached to this assignment.

Note: You will also peer review another group's report. The submission system will assign you to review another group's work. Please follow the instructions given by the system to submit your own report and review another group's report. *You must return to this assignment after the first submission deadline to receive another group's document to review.*

Submission: File submission. Upload the group's document as a PDF.

Grading: The assignment is graded with max. 7 points. The overall grading criterion is whether the report can aid in improving the quality of the SUT for the chosen stakeholders and whether it reflects an understanding of software quality assurance. The grading takes into account the chosen stakeholders, the appropriateness and clarity of their identified needs; the level of relevant detail and rationale given for the quality control activities; the clarity and precision of the test plan; the appropriateness and clarity of the information in the test report; the appropriateness of the conclusions drawn regarding the quality of the SUT; and the overall clarity, precision, informativeness, traceability, and appropriateness of the text.

Use of AI: You may use AI tools to get inspiration for your submission, but you must not directly use AI-generated text in the answer. This is a group assignment and collaboration among the members of your group is required. You may also discuss the assignment with members from other groups, but your group's submission must not include parts made by other groups.

Final report (points and feedback)

Final report peer review (points)

◄ **Previous section** Individual assignments

Next section ► Materials